

MEET BANTHIA

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Education

New York University - Courant Institute of Mathematical Sciences

Aug 2025 – May 2027

Master of Science in Computer Science

GPA : 3.93/4.0

National Institute of Technology Karnataka, Surathkal

Dec 2021 – May 2025

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.51/10

Experience

NYU MLSys Lab - Research Assistant under Prof. Jinyang Li

Dec 2025 – Present

- Currently extending a Vulkan-based 3D Gaussian Ray Tracing framework to implement and evaluate rendering-time optimizations, leveraging low-level control over rendering kernels and GPU resources to analyze performance bottlenecks and improve execution efficiency.
- Led research in 3D Gaussian Ray Tracing, focusing on systems-level optimization, latency-hiding techniques, and bounding volume hierarchy (BVH) design. Profiled and analyzed CUDA kernels, RT-core utilization, and workload characteristics to identify performance bottlenecks, and proposed and evaluated optimization strategies for improving rendering performance.
- Studied NeRF and 3D Gaussian Splatting systems (**3DGS**, **TC-GS**, **CLM**), profiling Tensor-Core-based CUDA kernels to identify performance bottlenecks, reproducing and validating CLM-GS results, and analyzing the **codebase** for correctness and consistency with reported performance.

Oracle - Server Technology Intern

May 2024 – Jul 2024

- Designed and implemented a Continuous Integration/Continuous Deployment (CI/CD) pipeline serving as the quality gatekeeper for Cloud Platform Services. And, configured automated triggers for service releases from active branches.
- Integrated automated test execution on Preflight (Gamma stage) environments where services were deployed in real-time.
- Established a 95% minimum pass rate threshold for integrated tests before upstream promotion to Fusion Apps.

Projects

ForkAndMove : Parallel Chess Engine[\[GitHub\]](#)

Sept 2025 – Dec 2025

- Built a high-performance chess engine using bitmap-based board representation for fast move generation and evaluation.
- Implemented an optimized parallel Alpha-Beta search and integrated Principal Variation Search (PVS) to improve pruning. Developed move generators and handled complex rules like castling and en-passant using bitwise operations.
- Analyzed performance improvements through advanced move ordering heuristics and algorithmic variations like multi-child PVS search.

ParJSON : Parallel JSON Parser in OpenMP[\[GitHub\]](#)

Feb 2026 – April 2026

- Designed and implemented a parallel JSON parser using OpenMP, addressing chunk-boundary dependencies arising from string-state tracking and escaped-character sequences in large JSON documents.
- Developed an optimized chunk-based parsing algorithm that reduced redundant memory writes and state-storage overhead through compact state encoding, improving scalability and execution efficiency on multicore systems.
- Evaluated multiple parallelization strategies and performed scalability analysis across varying thread counts and input sizes, achieving up to 4.7× speedup on large JSON files while maintaining correctness against Python's JSON parser.

Technical Skills

Languages: C/C++, Python, Standard ML, Elixir

Systems and Performance: Linux, Bash, CUDA, Vulkan, OptiX, Nsight Systems, GPU Profiling, strace, Slurm, OpenMP

ML Toolkit : Scikit-learn, Pandas, NLTK, LexNLP, Keras, TensorFlow, PyTorch, NumPy, OpenCV

Relevant Coursework: Distributed Systems, Multicore Processors, Programming Parallel Algorithms, ML Systems, Honors Analysis of Algorithms, Computer Networks, Wireless Networks, Operating Systems, Database Management System, Computer Architecture, Advanced Data Structures, Discrete Mathematics, Approximation Algorithms

Volunteering

IEEE NITK - Executive Member, CompSoc Group

Nov 2022 – May 2025

- Organized and managed multiple technical events and hackathons including **Silicon Maze**, **Eureka**, **Praelium**

Web Enthusiasts' Club, NITK — Executive Member, Algorithm Group

Nov 2022 – May 2025

- Organized and managed technical events and hackathons like **CPLLeague**

Achievements

- Secured **All India Rank 1117** in JEE MAINS among more than 1.2M+ candidates.
- Secured **All India Rank 2697** in JEE ADVANCE among 200K+ candidates